HavenOS 3.1 builder prompt: polish, personality and coordination

For the builder AI. HavenOS 3.0 (versionCode 12) to 3.1. Written 30 October 2026.

0. Ground rules

Project. HavenOS is an Android launcher for tablets and phones: package `app.havenos.launcher`, Kotlin, Jetpack Compose, minSdk 26, own code MIT licensed. Primary test device: Redmi Pad 2, tablet, landscape. It is a launcher only, never an operating system, and it replaces nothing on the lock screen.

Inspect before you change anything. Read the README and docs (including the HavenOS How It Works manual if it is in the repo), the module and package layout, the Gradle files and dependency catalog, the four states (`LayoutState`, `SettingsState`, `LookTokens`, `UiState`) and their single writer, the twelve event flows, the Settings panel, the mascot code, the online-API layer (one HTTP client, host allowlist, cache table, workers) and the existing tests. Run assemble, lint and unit tests once and record the baseline, including failures that already existed. Continue in the existing project. Do not scaffold a new one and do not rewrite working architecture.

Preserve, no exceptions.

- One writer per state: the interface sends events, only the state layer writes, one drag or drop is one atomic background write, slider writes stay debounced at 200 ms.
- Additive database migrations only. Existing layouts, backups and snapshots must still load, and old backup versions must still import.
- The four-zone shell (pager, dock, page indicator and edit toolbar space never overlap), 48 dp touch targets, Reduce Motion always wins, the Before / Changes / Undo model, and the twelve existing event flows.
- Privacy stance: no ads, accounts, analytics, telemetry or cloud sync. Network use stays limited to weather plus the toggled online APIs. This prompt adds no cloud AI and no new network calls.
- Permissions are asked only at the point of use, with a one-sentence reason first.

One freeze is lifted. Earlier work froze the Settings panel layout, hubs, rows, copy and order. That freeze is lifted for visual styling and for the specific changes listed in Phase C. Hubs, row order and copy stay as they are unless a Phase C item says otherwise.

How to work.

- Do phases A to G in order. Phase B (design system) must be finished before C and E use it. After each phase: build, run lint and tests, fix what you broke, write a short phase report, and make one commit.
- Add tests with the code. Prefer pure functions for logic (mappers, classifiers, planners, contrast math) so they are unit-testable without a device.
- Before adding a dependency, check whether the project already provides an equivalent. List every dependency you add and why. Prefer none.
- If this prompt contradicts the code, trust the code, say so in the report, and adapt with the smallest change.
- Anything marked as a hypothesis must be confirmed in the code before you act on it.

Honesty rules. Report only checks you actually ran. If there is no device or emulator, say that visual and gesture behavior is built but not proven, and list it under unverified. Never invent API shapes, model download URLs, checksums, benchmark numbers or test results. Do not publish, upload, change signing, or touch production data. In the final commit set `versionName` to 3.1.0 and `versionCode` to 13.

Evidence. The symptoms below come from two screen recordings on the Redmi Pad 2 (24 and 30 October 2026) and the HavenOS 3.0 manual. If you cannot view the recordings, rely on the exact symptom text given here.

1. Intent

The home screen is the part people love: glass cards over the wallpaper, big light numerals, small accent labels, a floating dock. It stays, with small fixes. Everything else (Settings, drawer and search, Today online cards, menus, dialogs, mascot) looks basic: flat gray cards, color emoji as icons, bare controls, truncated labels and a harsh set of six accent colors. Goals, in order:

1. Fix data, coordination and edge bugs (Phase A).
2. Build one design system, Haven Glass, and a flexible accent system (Phase B).
3. Apply it to every non-home surface (Phase C) and make the small home fixes (Phase D).
4. Rebuild the mascot as pixel art that walks on ledges, and make Find reliable (Phase E).
5. Lay the foundation for private, on-device AI (Phase F).
6. Verify honestly and update the docs (Phase G).

Phase A: fix data, coordination and edge bugs

For each item: confirm the cause in code, fix it, add tests.

A1. Scores card shows null and raw statuses.

- Symptom: rows like `Hayleys Group A null – null Hnb`, `Hong Kong null – null England`, `Incredibles null – null Warriors`, a row `South Africa - – - Oman`, and raw statuses such as `Abnormal` and `Delayed`.
- Cause (hypothesis): nullable JSON fields are string-interpolated and status codes are shown as received.
- Required:
  - Parse every score and team field as nullable. Never render the text null or a placeholder dash pair.
  - One display model per row: teams on the left, result or time on the right, competition as a muted second line.
  - Not started: show `vs` and the local start time if known. Live or finished: show scores in the provider's cricket or football format. Missing scores on a live or finished match: show the status text only.
  - Map provider status codes to plain labels in one table (for example Not started, Live, Finished, Delayed). A code the table does not know shows no status text. Do not guess what Abnormal means: check the API notes in the repo or the provider documentation and map it only if the meaning is documented.
  - Drop a row if both team names are missing.
  - Keep the Powered by SportScore attribution in every state.
- Acceptance: unit tests with fixtures for null scores, one null team, both teams null, an unknown status, a live match, a finished match and an upcoming match. No UI string can contain the word null.

A2. Wrong offline state when the device is online.

- Symptom: Wi-Fi is on and Scores loaded, but Quote of the day says `Offline - no saved data yet` and the NASA card says `No saved picture yet - check your connection and try again later`.
- Cause (hypothesis): the typed HTTP result (success, offline, rate-limited, not-found, other) collapses into one offline state before it reaches the card, or parse failures are mapped to offline. The quote endpoint's live shape was never confirmed, and the NASA demo key is shared and can return HTTP 429.
- Required:
  - Classify results in one place. Offline only when the device has no validated internet network (check the connectivity manager's network capabilities). NetworkError when the network is up but the host was unreachable. RateLimited for HTTP 429. NotFound for 404. ParseError when the body does not match the expected shape. ServerError for 5xx. Other otherwise.

  - Card states by class: Offline with cache shows the cached content plus `Updated HH:MM`. Offline without cache shows `Offline. Nothing saved yet.` RateLimited shows cached content if any plus `Busy right now. Try again later.` Every other class shows `Couldn't load. Retry.` with a working Retry action. The phrase check your connection may appear only for the Offline class.
  - Keep stale cache on failure with the offline or stale flag, as before. No retry storms: workers still return success without fetching when the toggle is off, and failures still return success.
  - Add a Diagnostics row per API in Device and Data: last result class, HTTP code, time, and the first 200 characters of the response body for parse errors. Never log keys, tokens or personal data. This lets the developer confirm the quote and NASA response shapes on a real network.
- Acceptance: unit tests mapping each result class to the right card state, and a test that Offline is not produced when the network capability is present.

A3. Crypto card nags when a key is missing.

- Symptom: the Crypto toggle is On with no CoinGecko key, and the card permanently shows `Add your free CoinGecko Demo key in Settings -> APIs to see prices.`
- Required: toggle Off means no card and no fetch. Toggle On without a key shows a compact key-needed state with one button, `Add key`, that deep-links to the CoinGecko key field in the APIs hub (opens Settings at APIs, scrolls to the field, focuses it). Toggle On with a key shows prices. Crypto stays default Off for new installs. Never fetch without a key.
- Acceptance: three state tests plus a deep-link test.

A4. The next page leaks into the current page.

- Symptom: on the first home page a sliver of the next page shows at the right edge (partial colored icon circles and the text fragment `Fri, 3` from the next page's glance row).
- Cause (hypothesis): page content is narrower than the pager viewport, or pages are not clipped to their bounds.
- Required: each page's width equals the pager viewport width after the Today rail and insets are subtracted. Clip page content to page bounds. No neighbor page content is visible at rest. Check portrait, landscape, Today rail on and off, and with Settings open.
- Acceptance: a layout test asserting page width equals viewport width, plus a manual tablet check.

A5. The glance row draws while the Today rail is visible.

- Symptom: a faint line such as `Fri, 30 Oct - 68F Overcast` draws at the top of the workspace while the Today rail already shows that information.
- Required: the glance row is composed only when Today is not visible (the rule in the manual). Remove any duplicate or ghost instance, including the one that belongs to the next page.
- Acceptance: a test that glance visibility equals not-Today-visible for each size class.

A6. The Today rail runs under the dock.

- Symptom: the last visible Today card is clipped flat where the dock begins, so its bottom edge and corner radius are cut.
- Required: the rail's scroll content has a bottom inset equal to the dock height plus 16 dp (the matching side inset for a left or right dock) so the last card scrolls fully clear and shows its bottom edge.
- Acceptance: a layout test for the inset value plus a manual tablet check.

A7. Dock Search and Library icons look like stray glyphs.

- Symptom: the dock's Search shows a tiny Q-like glyph and Library a bare 2x2 grid, unlike the app icons beside them.
- Required: both use proper icon tiles the same size as the dock's app icons (never below 48 dp), in the Haven Glass tile style from Phase B, with content descriptions Search and Library.
- Acceptance: the dock capacity rules (about 72 dp per slot) are unchanged and still tested.

A8. Ghost text during transitions.

- Symptom: during the APIs hub transition both the old and new rows draw on top of each other. The selected Milo pill in the Mascot hub shows a doubled label. The detail header shows a dark ghost Look behind the Back control.
- Cause (hypothesis): a crossfade that draws both contents at full size, and a selection indicator that animates under a label that is drawn twice.
- Required: hub transitions use one shared-axis transition (220 ms; 100 ms fade under Reduce Motion) where the outgoing content is removed from composition when it finishes and never overlaps the incoming layout. Segmented controls draw each label exactly once, above the sliding thumb. The header draws one title.
- Acceptance: a screenshot or composition test if a harness exists, otherwise a manual item.

A9. Two undo systems and a precedence contradiction.

- Problems: the toolbar Undo (50 levels, layout) does not undo Settings values, which have only Before / Changes. And choosing a density clears grid overrides although per-item should beat global.
- Required:
  - Settings value changes made while edit mode is open are recorded on the same undo timeline as layout changes, so the toolbar Undo and Redo include them, and the Changes list still shows them with their own Revert. Outside edit mode, the Changes list stays the way to revert.
  - Undo and Redo expose what they will undo in the accessibility label, for example Undo icon size change.
  - Precedence stays per-item over per-page over per-profile over global. Changing density must not delete per-item or per-page overrides silently. If overrides exist, show a confirmation: Density will reset N custom sizes. Keep them or reset? with Keep and Reset buttons. Keep is the default, and undo restores whatever was reset.
- Acceptance: tests for the shared timeline and for the density-with-overrides prompt.

A10. Long-press menu names do not match the hubs.

- Symptom: long-pressing the home shows Wallpaper, Layout, Widgets, Dock, Style, Home settings, Edit home. Names differ from the hubs (Look, Home and Dock) and the menu is not described in the manual.
- Required: each item opens exactly one destination and its label equals the destination's name. Wallpaper opens the wallpaper picker. Layout opens Home and Dock at the grid section. Widgets opens the add-widget sheet. Dock opens Home and Dock at the dock section. Look (rename Style) opens the Look hub. Home settings opens Home and Dock. Edit home enters edit mode. Document this entry in the manual and treat it as event flow 13 (see A11).
- Acceptance: a test that every menu action maps to a destination and a label.

A11. New event flows. The twelve flows predate the online cards, mascot and island. Add four named flows, each a pure function or a named host call with tests, in the same style as the existing ones:

- 13. Long-press menu action to destination (A10).
- 14. Mascot Find: query, target resolution, navigation, pointing, cancel (Phase E).
- 15. Island feeds: battery, notification and media observers to the pill's state and priority (Notification, then Media, then Charging; a notification is transient for 6 seconds).
- 16. Online API toggle or key change: toggle or key edit, worker enabled or disabled, card state, Diagnostics row. Document them in the manual and state which component owns each state they touch.

A12. Performance. Hypotheses to check and fix, then measure if a device is available:

- The Settings live preview re-lays out the home and recomputes per-cell wallpaper legibility on every debounced tick. Cache the downsampled luminance map once per wallpaper. On grid or icon-size changes recompute only the per-cell aggregation. Use stable keys and derived state so one setting change does not recompose the whole home.
- The island polls media state every 2 seconds. Use media session and controller callbacks through the existing notification listener, with no polling loop.
- Seven unconstrained periodic workers: keep one unique job per API, but make sure a disabled API's worker is cancelled, not just a no-op. Xiaomi battery management can defer background work, so a card must also refresh on open when its cache is stale.
- Add a baseline profile if the project has none, and macrobenchmark scripts for cold start and Settings open only if the project already uses macrobenchmark. Do not report numbers you did not measure.

A13. Safety.

- If the navigation bar is hidden through the secure-settings write, the previous value is stored. On every launch, if the bar is hidden but the user setting says it should not be, restore it. Add a Restore navigation bar row in Gestures and Motion. Make sure a crash or uninstall cannot leave the user without a bar.
- The gesture layer depends on an accessibility service the system may stop. Detect permission granted but service not connected and show the Needs-attention banner (`Gestures are paused. Reconnect`) with a button that opens the right system screen.

Phase B: the Haven Glass design system and the accent system

B1. One entry point. All screens other than home get their look from one module (for example `ui/glass`) that exposes `Modifier.glass(level)`, the components below and the `HavenTheme` tokens. Reuse the frosted implementation the home and dock already use. Do not invent a second blur approach. No screen may build its own surface color, radius or shadow.

B2. Materials.

- Glass1 (cards, widget frames, grouped lists): dark mode tint white 7% and hairline white 12%; light mode tint black 5% and hairline black 10%. Radius 28 dp for cards (matches the home), 20 dp for grouped lists.
- Glass2 (panels, sheets, drawer): tint white 12% in dark and white 70% in light, hairline 16%, 28 dp top corners for sheets.
- Glass3 (menus, popovers, dialogs, speech bubbles): tint white 16% in dark, hairline 18%, radius 20 dp for menus and 28 dp for dialogs.

- Depth comes from blur, tint and hairline, not shadows.
- Legibility guarantee: any glass surface that carries text sits on at least a 40% dark scrim in dark mode (60% light scrim in light mode) so text keeps 4.5:1 contrast on any wallpaper. Test with a very bright wallpaper and a busy wallpaper.
- Fallback when Reduce Effects is on, blur is unavailable (below API 31 or a device that disables it) or frame time suffers: same layout, no blur, tint alpha plus 6 points. Never fully transparent text backgrounds.
- At most two blurred layers on screen at once. A third becomes a dialog or uses the fallback.

B3. Shape, spacing, type.

- Radii: card 28, group 20, control 14, chip full, icon tile 12. Spacing scale 4, 8, 12, 16, 24, 32 dp. Row height 56 dp minimum (it grows for two lines). Touch targets 48 dp.
- Type scale (sp): numerals display 56 light (the home clock style), hero numerals 34 light (hub hero values and big readouts), title 28 regular, section label 12 medium with 0.6 sp letter spacing in the accent color, row label 16, row value 14 muted, caption 12 muted. Line height about 1.35.
- The signature is light large numerals plus small accent labels. Use both outside the home (hub hero values, battery, scores).
- Labels wrap to two lines before they ellipsize, and ellipsized text exposes its full text on long-press. Nothing in Settings may truncate at font scale 1.0 on the tablet. Examples to fix: `Quote of the ...`, `NASA pictur...`, `GitHub activi...`, `Currency rat...`, `Reduce moti...`.
- Honor system font scale up to 1.3 without clipped or overlapping text.

B4. Icons.

- One outline icon family (Material Symbols Rounded, outlined, weight 400, as vector drawables, or the extended icon set if it is already a dependency). Map every hub, row, permission, dock and menu entry through one `HavenIcon` enum.
- Icon tile: 36 dp, 12 dp radius, accent soft background (strength-dependent), accent-colored icon.
- Remove every emoji from the UI (the rail, hub tiles, permission rows and anywhere else). Add a unit test that scans string resources and Kotlin string literals under `src/main` for emoji and pictograph code points (for example U+1F300 to U+1FAFF, U+2600 to U+27BF, U+2B50, U+FE0F) and fails the build when it finds one.

B5. Components (each with pressed, focused and disabled-with-reason states and TalkBack roles).

- `GlassCard`; `GroupedSection` (a section label plus rows in one Glass1 container with hairline dividers, not a card per row); `SettingRow` variants: navigation, switch, segmented, slider, value, action, info.
- `HavenSegmented`: recessed track (white 6% in dark), sliding accent-soft thumb with accent text (spring), each label drawn once.
- `HavenSwitch`: 52 x 32 dp, a check mark on the thumb when on, accent track when on, neutral track when off.
- `HavenSlider`: no minus and plus buttons; a value bubble while dragging; a haptic tick at each step; an Auto chip where the setting has an Auto mode; writes debounced as before.
- `HavenBanner`: tonal accent-container background, icon tile, one action, no hardcoded colors.
- `HavenDialog`, `HavenMenu`, `HavenSnackbar` (Glass3); `EmptyState` (a small mascot pose, one-line headline, one-line body, one verb button); `Skeleton` (shimmer, static under Reduce Motion).
- Help: replace the tiny info glyph with a 48 dp touch target that opens a Glass3 popover with the help text.

B6. Motion. Durations stay: app open 320 ms, folders 280 ms, dock 240 ms. New hub transitions are 220 ms shared axis. Switch and segmented controls use one spring. Haptic ticks on switch, step and snap. Motion speed (Off, Fast, Normal, Slow) scales all of them. Reduce Motion turns everything into 100 ms fades. Reduce Effects flattens glass to a translucent tint.

B7. Accent system. Replace the six swatches and the hex field.

- Sources (a segmented choice in Look, Accent): Wallpaper, System, Curated, Custom. Existing installs keep their current accent (migrated). New installs default to Curated, Ember.
  - Wallpaper: read dominant colors with `WallpaperManager.getWallpaperColors` (API 27 and up), which works for the system wallpaper even where the bitmap is unreadable, and use the existing palette extraction for HavenOS wallpapers. It stays a suggestion chip (`Match wallpaper`) and is never applied automatically. Hypothesis to verify on the tablet: it works for the system wallpaper on Android 13 and up.
  - System: Material You dynamic color on Android 12 and up (`dynamicDarkColorScheme`, `dynamicLightColorScheme`), hidden on older versions.
  - Curated: twelve muted colors tuned for dark glass. Starting values (tune on the device, keep the names): Ember #F0A35E, Clay #D98C7A, Rose #E58AA6, Orchid #C490E4, Iris #9B8CFF, Cobalt #6FA0FF, Sky #7CC4F2, Lagoon #5CCBC0, Sage

#8FC9A0, Moss #B5CF7A, Sand #E3CFA0, Mono #E8E8EA.
    - Custom: a hue strip plus a lightness and saturation square, a live preview on the Look hero, a visible `Adjusted for readability` note when the color was clamped, up to six recent colors, and an eyedropper that samples the wallpaper preview. Hex entry moves under Advanced.
- One seed color generates every accent role through one function: accent (text and icons on glass), onAccent (text on accent fills), accentContainer (soft fills, alpha set by strength), accentOutline, for both dark and light modes. Use Material Color Utilities (HCT tones) if it is already a dependency, otherwise do the tone mapping in a perceptual space with a contrast clamp and add no dependency. Components read only these roles. Add a lint or unit test that fails on hex color literals in composables outside the token module (the purple Default launcher card is one known offender).
- Contrast rules enforced in code: accent text on glass at least 4.5:1, accent icons and borders at least 3:1, onAccent on accent at least 4.5:1. A seed that fails is clamped, never rejected.
- Accent strength: Subtle, Balanced, Vivid (default Subtle). It sets how much accent tints surfaces (soft container alpha about 10%, 18%, 30%) versus only highlighting state. Accent marks state (selected, on, primary action) and small section labels. It must not fill large areas by default.
- Migration: keep every stored custom color, map the old six swatches to the nearest curated color, and store names with values. The Changes list shows names (for example Accent: Iris to Ember), never raw integers (the current list shows values like -1717893).
- Acceptance: a unit test that every curated color passes the contrast rules in dark and light, a migration test, and a test that role generation is deterministic.

Phase C: apply Haven Glass to every non-home surface

Do these in order: C1 Settings, C2 Drawer and Search, C3 Today online cards, C4 Menus and dialogs, C5 Edit toolbar and context bar, C6 Page overview, widget picker and folders, C7 Calm, C8 Island.

C1. Settings (styling per Phase B plus the listed changes).

- Panel: Glass2. The tablet keeps master-detail (rail left, detail right, home re-laid in the free area as now). The phone keeps the bottom sheet with peek, half and full positions.
- Rail: 260 dp wide on tablet, rows 48 dp, an outline icon (24 dp) in an icon tile. The selected row is an accent-soft pill with accent icon and label. Labels wrap to two lines instead of truncating.

- Header: a back chevron and one 28 sp title. Remove the repeated `Settings > Settings` crumb and show a crumb only at depth three or more (for example `Look > Wallpaper`). `Before` becomes a labeled pill with an eye icon (accessibility label: Hold to see your home before changes). `Changes (N)` is a second pill. Close stays.
- Hub screen: a filled glass search field (copy unchanged: Search settings or try 'bigger icons'); the needs-attention banner as `HavenBanner`; a hero card that is a live miniature of the current look (accent, mode, material, icon shape, grid) built from tokens only, not a scaled-down copy of the real home. If that costs more than a trivial frame budget, fall back to the existing Minimal card, restyled. Quick tune as one grouped section, then recent chips, then the hub list as one grouped section with icon tiles, titles, live summaries and chevrons.
- Look hub: the accent Source choice and curated swatch grid (names under the swatches, a selected ring), the strength control, mode, material, label mode, icon shape previews, wallpaper source and style, all per Phase B.
- Home and Dock: grid steppers and density chips. Fix the Follows density chip so it never wraps letter by letter (give it a minimum width or move it to a caption line). The grid-shrink preview becomes a grouped section with the move summary written as one sentence plus an expandable list.
- APIs: each row has the API name (two lines allowed), a status line (`Updated 10:05`, `Needs key`, `Off`, or the A2 error text) and a switch. Key fields (CoinGecko, NASA, GitHub username) sit in a grouped section with Paste, Show and Hide, and validation.
- Gestures and Motion: restyle the device diagram with tokens (no hardcoded purple); toggles use `HavenSwitch`; the Test buttons still fire the real dispatcher.
- Device and Data: permission rows with an icon tile, a status chip (Allowed, Not allowed) and Manage. The default-launcher banner uses `HavenBanner`. Diagnostics gains the A2 rows and the A13 restore row if it lives here.
- Privacy: gains the AI Privacy Ledger entry (Phase F).
- Mascot: see Phase E4.
- Every hub shows loading, empty and error states where its data can be missing. Settings search results use the same rows and a `No settings match` empty state.
- Acceptance: no emoji anywhere; no truncated label at font scale 1.0 and 1.3 on the tablet; all contrast rules pass; TalkBack reads each row as label, value, role; hub transitions show no ghost frames.

C2. Drawer, Library and Search.

- The drawer is a Glass2 sheet with the legibility scrim, so home text never shows through (the 30 October recording showed `Sort: A-Z` overlapping the Agenda card behind the drawer).

- Keyboard-aware: use IME insets. When the keyboard opens, the search field stays visible, the results list resizes, and nothing overlaps.
- Sort and view options move into one compact toolbar row (icon buttons with a menu) so they cannot collide with other text.
- All, Categories and List become a `HavenSegmented`. The A-Z rail keeps its 48 dp hit area and gets a Glass3 letter bubble.
- Search: a Glass2 field. Results are grouped under section labels (Apps, Contacts, Settings, Calculator, Currency, Dictionary, Web) with icon tiles and match highlighting. The empty state uses the sad mascot pose. The keyboard's Search action runs the search.
- Acceptance: screenshots with the keyboard open and closed. Ranking, sources and toggles are unchanged.

C3. Today online cards.

- One card anatomy: header (small accent label, optional icon, overflow), body, footer line (`Updated 10:05 - Source`, caption muted).
- States come from A2 and A3, with `Skeleton` while loading.
- Scores: a Cricket and Football `HavenSegmented`; rows from A1, at most 4 plus `Show more`.
- Quote: 18 sp regular text, author in caption.
- Astronomy picture: rounded thumbnail, title, and `Set as HavenOS wallpaper` (HavenOS wallpapers only, as before).
- Crypto: symbol, price, and 24-hour change with an arrow icon and a sign (never color alone).
- GitHub, Currency and Dictionary use the same anatomy.
- Card order stays as in the manual. Offline, error and empty states use a static mascot pose (E4).
- Acceptance: a preview of every state in dark and light.

C4. Menus, dialogs, bubbles. Use `HavenMenu` and `HavenDialog` everywhere: icon plus label rows of 48 dp, destructive entries tinted with the danger color and text, dialogs with a title, a body, a text button and at most one filled button. The long-press menu becomes a Glass3 pill row with icons (A10). Snackbars become `HavenSnackbar` with Undo.

C5. Edit toolbar and context bar. Keep the 56 dp toolbar, its reserved space and every action. Restyle as a Glass3 pill: icon plus label, the active tool in accent-soft, hairline dividers between the Left, Center and Right groups, overflow unchanged. The context bar uses the same style. The selection ring stays 2 dp in the accent.

C6. Page overview, widget picker, folders, page indicator, app info. Glass2 sheets and Glass1 thumbnails (rounded 20 dp). The picker gets a search field plus size chips; previews are unchanged. The indicator keeps 8 dp dots and the 24 dp accent dash.

C7. Calm. The same system in a calmer variant: larger type, lower chrome contrast, accent used only for the schedule state. Exit still restores the home exactly.

C8. Island. Restyle with Glass3 and tokens. Behavior and priority are unchanged. It stays off by default and hidden in edit mode, drawer, search, Settings, quick panel and Calm.

Phase D: small home updates

Only these changes. Do not alter the home grid, icon sizes, spacing, the Today structure or the glass look.

- The home search widget becomes a Glass1 card matching the Today cards.
- Dock Search and Library tiles (A7).
- Verify A4, A5 and A6 on the home pages.
- The mascot never sits on an icon or on card content (Phase E).
- Everything else on the home is unchanged.

Phase E: the pixel mascot, rebuilt

E1. Art.

- Milo and Mila stay as the two characters (Milo squarer with a tuft, Mila rounder with a ribbon), redrawn as original pixel-art sprites. The style reference is classic 8-bit and 16-bit platformers. The characters must be original: no resemblance to any existing franchise character, no red cap, mustache or overalls, no borrowed sounds or assets.
- 24 x 24 px cells, transparent background, a 1 px dark outline, 8 colors per character: outline, base, shade, highlight, eye, blush, accent detail, white. The accent detail pixels (tuft tip, ribbon) are palette-swapped at load from the current accent seed, so the mascot follows the accent.
- Draw at an integer scale chosen from the size setting so the on-screen size is close to Small 56 dp, Medium 72 dp, Large 96 dp (scale = round(targetPx / 24), minimum 2). Use `FilterQuality.None` and no smoothing. Never use fractional scales.
- Both characters share one frame layout so code is shared: idle (4 frames, blink, 4 fps), walk (6 frames, 10 fps), jump (rise, apex, fall, land x2), sit (2), sleep (3 slow frames with a Z), wave (4), happy (4), surprised (2), sad (2), think (3, used while Find searches), point (2, with a small arrow), poof (4, used for Dismiss and for hiding). Face right and mirror for left.

- Deliver one PNG sheet per character plus a small JSON atlas (or palette-indexed maps in code if you cannot produce PNGs). Also commit a preview image under `docs/mascot/` showing all frames at 4x on a gray and on a dark background, so the art can be reviewed without running the app. If you cannot judge your own pixel art, say so. Do not claim it looks good.
- Keep the old vector mascot behind the same interface behind a debug flag until the new one is accepted. Remove it in a later change.

E2. Behavior engine.

- State machine: Hidden, Spawning, Idle, Walking, Jumping, Falling, Sitting, Sleeping, Reacting (tap), Carried (long-press drag), Pointing, Despawning. Transitions are pure functions with tests.
- Ledge graph, built from the live layout. Ledges are horizontal segments: the top edges of Today cards and widget frames, gutters between icon rows, the top of the dock, empty cells and free space. A ledge segment is valid only where the area above it, as tall as the sprite, is free of interactive content (icons, labels, widget hit areas). Cut or drop the parts that fail. So the mascot never stands on an icon or covers one.
- Edges: Walk along one ledge. Jump up to two icon-rows high between ledges whose horizontal ranges overlap or are within 3 cells. Drop to any lower ledge with a fall animation. No edges across pages while a page is swiping. The graph covers the visible page, the Today rail and the dock.
- Planner: pick a target ledge point (weighted toward the dock top and empty space, away from busy areas), walk or jump there at 40 to 60 dp per second, dwell 3 to 12 seconds, then pick again. After 60 to 120 seconds idle, sit. After 5 minutes, sleep. Use an injectable random source so tests are deterministic.
- Rebuild the graph on every layout change (LayoutState version), rotation or size-class change, page change, density or icon-size change, dock position change, keyboard inset change, Settings open or close, and edit mode enter or exit. If the current ledge disappears, fall to the nearest valid ledge. If none exists, poof and respawn on the dock top.
- Visibility: show only on home pages and the Today rail. Hide with a poof when any of these opens: edit mode, a drag, the drawer, search, Settings (except the Find pointing state, E3), the quick panel, Calm, or whenever the home is not the resumed screen. Pause all timers and the frame loop when the home is stopped. Reduce Motion: no roaming. The mascot sits on the dock top in a static idle pose and still reacts to taps. Battery saver: idle and sit only. Roaming off in Settings behaves like Reduce Motion for movement.

- Rendering and performance: a single overlay layer that is touch-transparent except for the sprite's hitbox and the bubble. Position and frame come from time-based animation state read in a draw or graphicsLayer lambda, not from Compose state written every frame. The frame loop runs only while walking, jumping or falling; idle and sleep use timers. Frame stepping is time-based so speed does not depend on frame rate. No per-frame allocations.
- Long-press and drag: a 350 ms long-press on the sprite picks it up (Carried, a dangling frame, a haptic tick). Dropping it on any valid ledge point places it. An invalid drop returns it to the previous ledge. A short tap is a tap. Taps and long presses on anything else pass through.

E3. Interaction and Find.

- Bubble: a speech bubble with a pixel-stepped border (2 px steps at the sprite's scale) and readable system-font text (14 sp or larger), so accessibility is not sacrificed to style. Four chips with icons and labels, each 48 dp: Find, Help, Suggest, Dismiss. The bubble anchors to the sprite, flips side to stay on screen, and respects IME insets. Close and back dismiss it.
- Find is one tracked flow (event flow 14): Idle, Asking, Searching (think pose), Choosing (up to 4 choices when the match is uncertain), Navigating, Pointing, Done or Cancelled. Cancel on any touch outside the bubble, on back, on a mode change, or after 20 seconds without progress. A query with no match says so plainly (`I can't find that`) and offers search. It never guesses.
- The Asking step uses a styled text field with the Search keyboard action (the 30 October recording showed a bare outlined box). It submits with the action key or a Find button and keeps the bubble above the keyboard.
- Resolution reuses the existing search index through the existing adapter. There is no second index. Ranking: exact, prefix, synonym, fuzzy.
- Target kinds and behavior:
  - App icon on the visible page: walk or jump to the nearest valid ledge by the icon, face it, Point, show the spotlight ring.
  - App icon on another page: animate the pager to that page, wait until it settles, rebuild the ledge graph, then walk and Point.
  - App inside a folder, or an app only in the drawer: show the bubble with a Launch chip (a ring cannot be drawn inside a closed folder). Do not open folders unasked.
  - Widget: Point at its frame.
  - Settings control: open Settings at the right hub, scroll the row into view, and Point with a ring on the row while the panel is open (the one exception to the hide rule). Ending Find closes nothing; the user closes Settings.

- Spotlight ring: accent, 2 dp, a soft pulse (static under Reduce Motion), a 40% dim around the target. A tap anywhere dismisses it.
- Help answers from the local help map (expand it to cover the new surfaces) and can ring the control it is talking about. Suggest: at most one per session, a 24-hour cooldown, quiet hours 22:00 to 07:00. Teach: one-time tutorials, with don't show again respected. Dismiss sticks across restarts until the user turns the mascot back on in Settings.
- Chat stays scripted. Keep the `MascotBrain` interface and `ScriptedBrain`. Phase F adds an optional local-model brain behind the same interface.

E4. Settings hub and empty states.

- The Mascot hub (C1) shows the sprite animating in a preview box, a Preview button that plays idle, walk, jump and wave, and these rows: on/off (default off), character, roaming, size (Small, Medium, Large), Find, tips frequency (Off, Rare, Normal), quiet hours with time pickers, and Reset mascot.
- Static poses are reused as empty-state hosts: sleeping for `No events today`, sad for an offline or error card, surprised for a Find with no match, thinking for loading. These are static images, not roaming sprites, so they never conflict with the overlay layer.

E5. Tests.

- Unit: state machine transitions; ledge-graph construction from fixture layouts (icon rows, a widget, the dock, a Today rail) including the clearance rule and a layout with no valid ledge; planner determinism with a seeded random source; Find target resolution for each target kind including no match and uncertain; the invalidation triggers.
- Manual on the tablet: see the Phase G checklist.

Phase F: on-device assist foundation

F0. Principle. Assist features work from data already on the device, run on the device, and never send screen-derived or usage data off it. This pass adds no cloud AI. Anything that needs a download is optional, user-initiated and visible.

F1. Seam. Define `AssistProvider` (turn a typed request into candidate actions) with one implementation, `LocalAssist` (rules, synonyms and ranking, no model). Keep `MascotBrain` as the chat seam. Do not add a cloud provider.

F2. Lite improvements (no model, no new dependency).

- Search and Settings search understand compound requests through a small command grammar, for example `dark mode and bigger icons`, `silent`, `open camera`, `go to Privacy`, resolved through the action router (F3). Unknown phrases fall back to normal search.

- App prediction stays on local launch history (count, recency, same-hour boost). Expose a Clear learned data control.
- Categories, suggestions and emotes remain rule-based.

F3. Action router. All assist and mascot actions are typed objects validated against a whitelist: open app, open a settings hub, set a setting (through the typed SettingsState accessors with range checks), set accent, toggle Calm, find a target. Resolve names against the real installed-app list and the real settings registry, and refuse anything not found. Actions that change settings show the preview and land in Before, Changes and Undo with readable names. Never allowed: uninstall, hide-all, permission grants, anything outside the whitelist. Confirm before anything destructive or hard to notice.

F4. AI Privacy Ledger (Privacy hub). A list of every assist feature (Search commands, App prediction, Mascot Find, Mascot chat, Suggestions) with what local data each reads, an On-device badge, the last used time, a per-feature switch, and Clear learned data. Entries record counts and times only, never content.

F5. Optional local model pack (scaffold only).

- Build the pieces a later local-model brain needs: a model-pack manager (download on user request, Wi-Fi only by default, progress and cancel, SHA-256 verification, app-private storage, a delete button, free-space and RAM checks), a `LocalModelBrain` that implements `MascotBrain` behind a feature flag, and a Settings row (Chat model, Not set up) that shows the real state.
- Do not bundle a model. Leave the download URL and checksum as empty placeholders, as the manual states, and do not invent values. Do not pick a runtime unless one integrates cleanly with the project. If you try one, keep it behind the flag and report what happened. A model must wake only when chat opens and must never change a setting without confirmation.
- Until a pack exists, the hub shows Chat model: Not set up and the scripted brain answers.

F6. Not in this pass: Scenes, Style DNA, Recall, a notification digest, cloud AI, mascot voice, more characters, an online themes store.

Phase G: verification, docs and report

G1. Checks. Run assemble debug and release, lint, unit tests, any instrumentation tests that can run, and any screenshot tests the project already has. Report each command and its result. Fix regressions you introduced and separate them from pre-existing failures.

G2. Docs. Update the HavenOS manual (sections 1 to 3, the event flows in section 4, the online card states in section 8, the mascot and island in section 9, the gaps in section 10) so it matches what actually shipped. Keep the honest gaps section accurate.

G3. Tablet checklist (mark each item not run unless it was run on the Redmi Pad 2).

- Home: no next-page sliver; no ghost glance row; the last Today card clears the dock; dock Search and Library tiles.
- Settings: each hub in dark, light and Reduce Effects; no emoji; no truncation at font scale 1.0 and 1.3; clean hub transitions; accent sources, curated colors, custom clamp and strength; the Changes list shows names.
- Cards: Scores with null data; Quote and NASA error classes with Wi-Fi on; Crypto in its three states; offline with cache.
- Drawer: keyboard open and closed, no overlap with home text.
- Mascot: turn it on; it roams without covering icons; rotation while walking; a page change; edit mode hides it; Settings hides it; Find an app on the same page, on another page, in a folder, a widget, a Settings control, and a no-match query; tap versus long-press-drag; Dismiss survives a restart; Reduce Motion.
- Safety: navigation bar restore; a stopped accessibility service shows the banner.
- Performance: scrolling the Today rail, opening Settings, dragging a slider. Note any jank.

G4. Final report. What changed per phase, tests added, checks run and results, dependencies added, files touched, assumptions, deviations from this prompt, unverified items, known gaps, and the next most useful improvement. Make no claim of production readiness, accessibility conformance or security review.